

# Nisha Bajiya

Computational Biologist - Artificial intelligence - Peptide and Aptamer Therapeutics

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## Profile

Computational biologist with a research focus on peptide and aptamer therapeutics. Experienced in building machine-learning predictors, manually curated databases, and public web servers for antibacterial peptides, cell-penetrating peptides, and anticancer, antiviral, and antibacterial aptamers. Proficient in feature engineering, deep learning, and protein and nucleotide language models (ESM-2, ProtBERT, DNABERT, Nucleotide Transformer), with software shared through GitHub and PyPI.

## Education

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|----------------------------------------------------------------------------|-----------------------|
| <b>Ph.D. in Computational Biology</b>                                      | 9.00/10 CGPA          |
| Indraprastha Institute of Information Technology Delhi (IIIT-Delhi), India | 2022 – 2027           |
| <b>M.Sc. in Biotechnology</b>                                              | 91.08% (2019)         |
| Jaipur National University, Jaipur, India                                  | University Gold Medal |
| <b>B.Sc. in Biotechnology</b>                                              | 82.7% (2017)          |
| University of Rajasthan (Biyani Girls College), Jaipur, India              | First Class           |

## Publications

Peer-reviewed journal

- **AptViralDB: A repository of experimentally validated antiviral aptamers**  
Bajiya N, Singh S, Gahlot PS, Raghava GPS Molecular Therapy - Nucleic Acids, 2026
- **Prediction of inhibitory peptides against *E. coli* with desired MIC value**  
Bajiya N, Kumar N, Raghava GPS Scientific Reports, 2025
- **CPPsite3: An updated large repository of experimentally validated cell-penetrating peptides**  
Bajiya N, Najrin S, Kumar P, Choudhury S, Tomer R, Raghava GPS Drug Discovery Today, 2025
- **AntiBP3: A method for predicting antibacterial peptides against gram-positive/negative/variable bacteria**  
Bajiya N, Choudhury S, Dhall A, Raghava GPS Antibiotics, 2024
- **Advances in the field of phage-based therapy with special emphasis on computational resources**  
Bajiya N, Dhall A, Aggarwal S, Raghava GPS Briefings in Bioinformatics, 2023
- **In-silico tool for predicting and scanning rheumatoid arthritis-inducing peptides in an antigen**  
Tomer R, Jain S, Gahlot PS, Bajiya N, Raghava GPS Frontiers in Immunology, 2025
- **Prediction of plant resistance proteins using alignment-based and alignment-free approaches**  
Gahlot PS, Choudhury S, Bajiya N, Kumar N, Raghava GPS Proteomics, 2024
- **Prediction of anti-freezing proteins from their evolutionary profile**  
Kumar N, Choudhury S, Bajiya N, Patiyal S, Raghava GPS Proteomics, 2024
- **MRSLpred: A hybrid approach for predicting multi-label subcellular localization of mRNA at the genome scale**  
Choudhury S, Bajiya N, Patiyal S, Raghava GPS Frontiers in Bioinformatics, 2024
- **Multi-perspectives and challenges in identifying B-cell epitopes**  
Kumar N, Bajiya N, Patiyal S, Raghava GPS Protein Science, 2023

## Preprints

- **AntiCapt: Fine-tuned nucleotide language models for predicting and designing anticancer aptamers**  
Bajiya N, Mehta NK, Raghava GPS bioRxiv, 2026
- **AptCancerDB: A curated knowledgebase and translational discovery platform for anticancer aptamers**  
Bajiya N, Singh S, Raghava GPS bioRxiv, 2026
- **AptBacterialDB: A comprehensive, manually curated database of antibacterial aptamers**  
Bajiya N, Gupta I, Raghava GPS bioRxiv, 2026
- **CPPLocPred: Subcellular localization of cell-penetrating peptides**  
Bajiya N, Mehta NK, Raghava GPS bioRxiv, 2026
- **Prediction of lncRNA-protein interacting pairs using LLM embeddings based on evolutionary information**  
Choudhury S, Bajiya N, Raghava GPS bioRxiv, 2025
- **In silico tool for identification of colorectal cancer from cell-free DNA biomarkers**  
Mathur K, Jain S, Bajiya N, Kumar N, Raghava GPS arXiv, 2025

## Book chapters

- **Chapter 19: Machine learning-based methods for designing protein-based drugs**  
Bajiya N, Rathore AS, Kumar N, Raghava GPS Royal Society of Chemistry, 2026, vol. 87, pp. 641–681
- **Browsing and mapping of cell-penetrating peptides in the CPPsite3 database**  
Bajiya N et al. Under publication
- **Prediction, design, and genome-wide scanning of plant disease resistance proteins using PlantDRPpred**  
Raghava GPS et al. Under publication

## Research Experience

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**January 2022 - Doctoral Researcher**, IIIT-Delhi, New Delhi

**January 2027** *Supervisor: Prof. Gajendra P. S. Raghava*

- Built ML/DL predictors for peptide activity and therapeutic aptamers
- Curated large sequence datasets from literature and public databases
- Developed Python pipelines on HPC systems for large-scale model training
- Applied peptide, protein, and nucleotide language models for feature extraction
- Released web servers, standalone tools, and GitHub/PyPI packages

**January 2019 - Dissertation Research (M.Sc.)**, Jaipur National University, Jaipur (Rajasthan)

**June 2019** *Supervisor: Dr. Divya Shrivastava, Dr. Smita Pareek*

- Dissertation on isolation and antifungal susceptibility testing of dermatophytes
- Processed clinical samples from 60 patients; KOH microscopy, SDA/DTM culture
- Identified species by LPCB staining and slide culture; disk diffusion testing

**June 2016 - Dissertation Trainee (B.Sc.)**, Dr. B. Lal Institute of Biotechnology, Jaipur (Rajasthan)

**August 2016** *Supervisor: Dr. Parul Chowdhury*

- Dissertation on drought-stress effects on antioxidant enzymes in *Vigna radiata*
- Pot experiment on two mung bean varieties under three drought levels
- Assayed CAT, SOD, APX, and GPX by UV-Vis spectrophotometry; Bradford assay
- Completed plant biotechnology training (1–15 June 2016)

June 2015 - Research Trainee, Biyani Girls College, Jaipur (Rajasthan)

July 2015 Supervisor: Ms. Heena Kewalramani

- Completed training and report on electrophoretic techniques (15 days)

## Current Projects

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### 1. Therapeutic Aptamer Prediction

### 2. Pan-Cancer Genomic Map of Therapeutic Aptamer Targets

## Skills

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|---------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Languages                             | Python, Bash/Linux, SQL (MySQL), HTML/CSS, JavaScript, PHP                                                                                                                                                               |
| Scientific Python and Cheminformatics | NumPy, Pandas, SciPy, Matplotlib, Jupyter Notebook, RDKit                                                                                                                                                                |
| Machine Learning                      | scikit-learn, XGBoost, random forest, TensorFlow/Keras, PyTorch, CNN, LSTM, cross-validation, feature selection                                                                                                          |
| Language Models                       | PepBERT, ProtBERT, BioBERT, ESM-2, DNABERT, HyenaDNA, Nucleotide Transformer                                                                                                                                             |
| Sequence and Structure Analysis       | BLAST(p/n)/PSI-BLAST, CD-HIT, MERCI motif analysis, Pfeature descriptors, Biopython, ViennaRNA                                                                                                                           |
| Data Resources                        | UniProt/Swiss-Prot, PDB, GenBank, GDC (Genomic Data Commons)                                                                                                                                                             |
| Cancer Omics                          | DNA methylation and gene expression in cancer cells                                                                                                                                                                      |
| Tool Development                      | Git/GitHub, PyPI packaging, standalone software, web servers built with MySQL and HTML/CSS/JavaScript, knowledge graphs, manual literature curation                                                                      |
| Laboratory                            | BSL-3 handling, bacterial and fungal culture maintenance, KOH and LPCB microscopy, antifungal disk diffusion, Bradford assay, antioxidant enzyme assays (CAT, SOD, APX, GPX), gel electrophoresis, plant pot experiments |
| Equipment                             | Bright field microscope, laminar air flow, UV-Vis spectrophotometer, refrigerated centrifuge, autoclave, hot air oven, pH meter, analytical balance, BOD incubator                                                       |

## Teaching and Mentorship

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Teaching Assistant, IIIT-Delhi, New Delhi

- Courses: Machine Learning for Biomedical Applications (two TAs); Big Data Mining in Healthcare (one TAs)
- Ran tutorial sessions on machine-learning methods, ways to assess models, and working with biological data
- Helped students build ML workflows and make sense of biological datasets
- Evaluated assignments and gave written feedback on students' computational work

### Student Mentorship

- Guided 3 students through their thesis research on computational biology and ML problems
- Supported data preprocessing, model building, validation, and database development
- Advised on project planning, debugging, interpretation of results, and workflow design

## Honors, Fellowships and Achievements

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- Dean's List, Best Teaching Assistant (BIO542), IIIT-Delhi, 2023–24
- CSIR-UGC NET Junior Research Fellowship, Life Sciences (2020): **All India Rank 97**
- ICMR Junior Research Fellowship, Life Sciences (2020)
- DBT Junior Research Fellowship (2020)
- GATE 2021, Biotechnology: All India Rank 334
- Qualified the Ph.D. entrance examination, All India Institute of Medical Sciences (AIIMS), New Delhi (2021)
- University Gold Medal for First Position, M.Sc. Biotechnology, Jaipur National University (2020)
- First Position, B.Sc. II Year (Biotechnology), Biyani Girls College, Jaipur (2015–16)

## Workshops, Symposia and Training

- Workshop: *Establishing and Sustaining Cloud Infrastructure for Informatics-based Healthcare*, IIIT-Delhi, 22–23 March 2024
- Symposium on Computational Gastronomy, IIIT-Delhi, 10 December 2023
- National Workshop-cum-Training Programme on *Open Source Tools in Bioinformatics*, Birla Institute of Scientific Research, Jaipur, 21–23 January 2015

## Extracurricular Activities

- Completed 5 km run at the Faridabad Half Marathon (3 March 2024) and 10 km at the Decathlon 10K Run, Delhi (22 September 2024)
- Second position in Yoga, Karwaan (IIIT-Delhi inter-branch league), 2025; performed at International Yoga Day (June 2026) and Happy Hours (March 2026), IIIT-Delhi
- Voluntary blood donation, IIIT-Delhi camp organized with Hindu Rao Hospital, March 2024